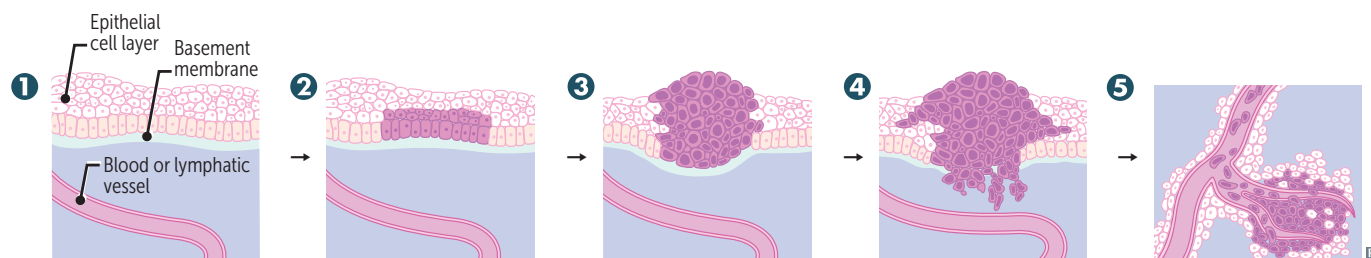


► PATHOLOGY—NEOPLASIA

Neoplasia and neoplastic progression

Uncontrolled, often monoclonal proliferation of cells. Can be benign or malignant. Any neoplastic growth has two components: parenchyma (neoplastic cells) and supporting stroma (non-neoplastic; eg, blood vessels, connective tissue).

**Normal cells**

➊ Normal cells with basal → apical polarity. See cervical example, which shows normal cells and spectrum of dysplasia, as discussed below.

Dysplasia

➋ Loss of uniformity in cell size and shape (pleomorphism); loss of tissue orientation; nuclear changes (eg, ↑ nuclear:cytoplasmic ratio); often reversible.

**Carcinoma in situ/
preinvasive**

➌ Irreversible severe dysplasia that involves the entire thickness of epithelium but does not penetrate the intact basement membrane.

Invasive carcinoma

➍ Cells have invaded basement membrane using collagenases and hydrolases (metalloproteinases). Cell-cell contacts lost by inactivation of E-cadherin.

Metastasis

➎ Spread to distant organ(s) via lymphatics or blood.

Tumor nomenclature

Carcinoma implies epithelial origin, whereas **sarcoma** denotes mesenchymal origin. Both terms generally imply malignancy.

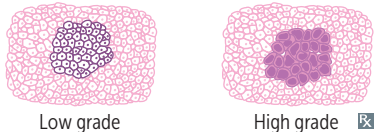
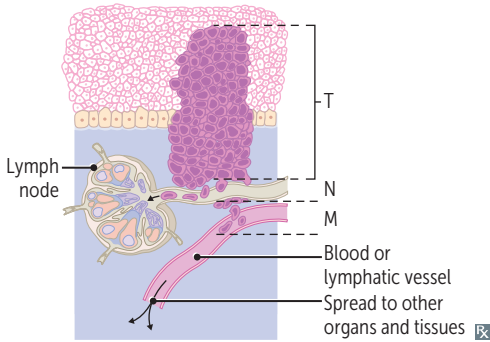
Benign tumors are usually well-differentiated and well-demarcated, with low mitotic activity, no metastases, and no necrosis.

Malignant tumors (cancers) may show poor differentiation, erratic growth, local invasion, metastasis, and ↓ apoptosis.

Terms for non-neoplastic malformations include hamartoma (disorganized overgrowth of tissues in their native location, eg, Peutz-Jeghers polyps) and choristoma (normal tissue in a foreign location, eg, gastric tissue located in distal ileum in Meckel diverticulum).

CELL TYPE	BENIGN	MALIGNANT
Epithelium	Adenoma, papilloma	Adenocarcinoma, papillary carcinoma
Mesenchyme		
Blood cells		Leukemia, lymphoma
Blood vessels	Hemangioma	Angiosarcoma
Smooth muscle	Leiomyoma	Leiomyosarcoma
Striated muscle	Rhabdomyoma	Rhabdomyosarcoma
Connective tissue	Fibroma	Fibrosarcoma
Bone	Osteoma	Osteosarcoma
Fat	Lipoma	Liposarcoma
Melanocyte	Nevus/mole	Melanoma

Tumor grade vs stage

Grade	Degree of cell differentiation (tissue of origin resemblance) and mitotic activity on histology. Ranges from low-grade (well differentiated) to high-grade (poorly differentiated or undifferentiated [anaplastic]). Higher grade often correlates with higher aggressiveness.	 Low grade High grade
Stage	Degree of invasion and spread from initial site. Depth of invasion correlates to risk of metastasis. Based on clinical (c) or pathologic (p) findings. TNM staging system (importance: M > N > T): - Primary tumor size/invasion. - Regional lymph node metastasis. - Distant metastasis. Stage generally has more prognostic value than grade (eg, a high-stage yet low-grade tumor is usually worse than a low-stage yet high-grade tumor). Stage (spread) determines survival.	 T N M Lymph node Blood or lymphatic vessel Spread to other organs and tissues

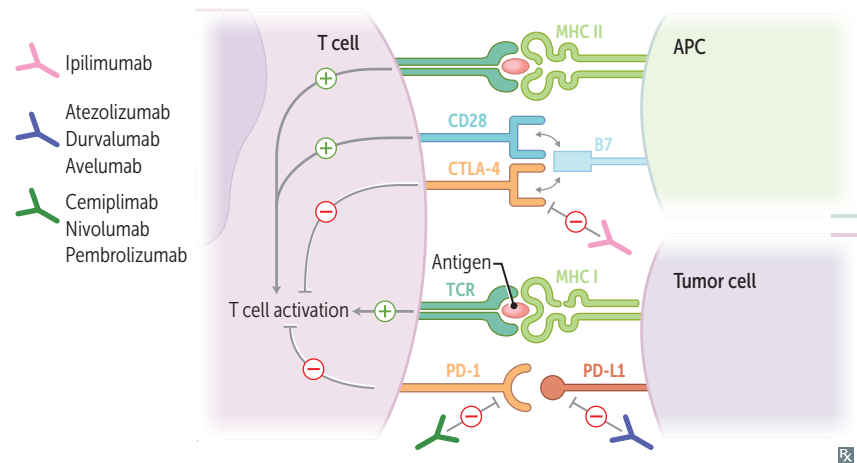
Hallmarks of cancer

Cancer is caused by (mostly acquired) DNA mutations that affect fundamental cellular processes (eg, growth, DNA repair, survival).

HALLMARK	MECHANISM
Growth signal self-sufficiency	Mutations in genes encoding: ▪ Proto-oncogenes → ↑ growth factors → autocrine loop (eg, ↑ PDGF in brain tumors) ▪ Growth factor receptors → constitutive signaling (eg, <i>HER2</i> in breast cancer) ▪ Signaling molecules (eg, <i>RAS</i>) ▪ Transcription factors (eg, <i>MYC</i>) ▪ Cell cycle regulators (eg, cyclins, CDKs)
Anti-growth signal insensitivity	▪ Mutations in tumor suppressor genes (eg, <i>Rb</i>) ▪ Loss of E-cadherin function → loss of contact inhibition (eg, <i>NF2</i> mutations)
Evasion of apoptosis	Mutations in genes that regulate apoptosis (eg, <i>TP53</i> , <i>BCL2</i> → follicular B cell lymphoma).
Limitless replicative potential	Reactivation of telomerase → maintenance and lengthening of telomeres → prevention of chromosome shortening and cell aging.
Sustained angiogenesis	↑ pro-angiogenic factors (eg, VEGF) or ↓ inhibitory factors. Factors may be produced by tumor or stromal cells. Vessels can sprout from existing capillaries (neoangiogenesis) or endothelial cells are recruited from bone marrow (vasculogenesis). Vessels may be leaky and/or dilated.
Warburg effect	Shift of glucose metabolism away from mitochondrial oxidative phosphorylation toward glycolysis, even in the presence of oxygen. Aerobic glycolysis provides rapidly dividing cancer cells with the carbon needed for synthesis of cellular structures, leading to ↑ lactic acid.
Immune evasion in cancer	Normally, immune cells can recognize and attack tumor cells. For successful tumorigenesis, tumor cells must evade the immune system. Multiple escape mechanisms exist: ▪ ↓ MHC class I expression by tumor cells → cytotoxic T cells are unable to recognize tumor cells. ▪ Tumor cells secrete immunosuppressive factors (eg, TGF-β) and recruit regulatory T cells to down regulate immune response. ▪ Tumor cells up regulate immune checkpoint molecules, which inhibit immune response.
Tissue invasion	Loss of E-cadherin function → loosening of intercellular junctions → metalloproteinases degrade basement membrane and ECM → cells attach to ECM proteins (eg, laminin, fibronectin) → cells migrate through degraded ECM (“locomotion”) → vascular dissemination.
Metastasis	Tumor cells or emboli spread via lymphatics or blood → adhesion to endothelium → extravasation and homing. Site of metastasis can be predicted by site of 1° tumor, as the target organ is often the first-encountered capillary bed. Some cancers show organ tropism (eg, lung cancers commonly metastasize to adrenals).

Immune checkpoint interactions

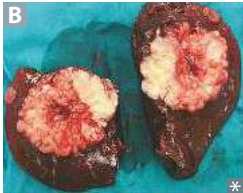
- Signals that modulate T-cell activation and function → ↓ immune response against tumor cells.
Targeted by several cancer immunotherapies. Examples:
- Interaction between PD-1 (on T cells) and PD-L1/2 (on tumor cells or immune cells in tumor microenvironment) → T-cell dysfunction (exhaustion). Inhibited by antibodies against PD-1 (eg, cemiplimab, nivolumab, pembrolizumab) or PD-L1 (eg, atezolizumab, durvalumab, avelumab).
 - CTLA-4 on T cells outcompetes CD28 for B7 on APCs → loss of T-cell costimulatory signal. Inhibited by antibodies against CTLA-4 (eg, ipilimumab).



Cancer epidemiology	Skin cancer (basal > squamous >> melanoma) is the most common cancer (not included below).			
	MALES	FEMALES	CHILDREN (AGE 0–14)	NOTES
Cancer incidence	1. Prostate 2. Lung 3. Colon/rectum	1. Breast 2. Lung 3. Colon/rectum	1. Leukemia 2. CNS 3. Neuroblastoma	Lung cancer incidence has ↓ in males, but has not changed significantly in females.
Cancer mortality	1. Lung 2. Prostate 3. Colon/rectum	1. Lung 2. Breast 3. Colon/rectum	1. Leukemia 2. CNS 3. Neuroblastoma	Cancer is the 2nd leading cause of death in the United States (heart disease is 1st).

Common metastases

Most **C**arcinomas spread via **L**ymphatics; most **S**arcomas spread **H**ematogenously (**CLaSH**). However, **four c**arcinomas **r**oute **h**ematogenously: **f**ollicular thyroid carcinoma, **c**horiocarcinoma, **r**enal cell carcinoma, and **h**epatocellular carcinoma. Metastasis to bone, liver, lung, and brain is more common than 1° malignancy in these organs. Metastases often appear as multiple lesions (vs 1° tumors which generally appear as solitary lesions).

SITE OF METASTASIS	1° TUMOR	NOTES
Bone 	Prostate, breast >> lung > kidney, colon	Predilection for axial skeleton A Bone metastasis can be: ▪ Blastic (eg, prostate, small cell lung cancer) ▪ Mixed (eg, breast) ▪ Lytic (eg, kidney, colon, non-small cell lung cancer)
Liver 	Colon > breast >> pancreas, lung, prostate	Scattered throughout liver parenchyma B
Lung	Colon, breast >> kidney, prostate	Typically involve both lungs
Brain	Lung > breast >> melanoma > colon, prostate	Usually seen at gray/white matter junction

Oncogenes

Gain of function mutation converts proto-oncogene (normal gene) to oncogene → ↑ cancer risk.
Requires damage to only **one** allele of a proto-**oncogene**.

GENE	GENE PRODUCT	ASSOCIATED NEOPLASM
ALK	Receptor tyrosine kinase	Lung adenocarcinoma
EGFR (ERBB1)	Receptor tyrosine kinase	Lung adenocarcinoma
HER2 (ERBB2)	Receptor tyrosine kinase	Breast and gastric carcinomas
RET	RE ceptor T yrosine kinase	MEN2A and 2B, medullary and papillary thyroid carcinoma, pheochromocytoma
BCR-ABL	Non-receptor tyrosine kinase	CML, ALL
JAK2	Non-receptor tyrosine kinase	Myeloproliferative neoplasms
BRAF	Serine/threonine kinase	Melanoma, non-Hodgkin lymphoma, colorectal carcinoma, papillary thyroid carcinoma, hairy cell leukemia
c-KIT	Cy to KI ne receptor (CD117)	Gastrointestinal stromal tumor (GIST), mastocytosis
MYCC (c-myc)	Transcription factor	Burkitt lymphoma
MYCN (N-myc)	Transcription factor	N euroblastoma
KRAS	RAS GTPase	Pancreatic, colorectal, lung, endometrial cancers
BCL-2	Antiapoptotic molecule (inhibits apoptosis)	Follicular and diffuse large B-Cell L ymphomas

Tumor suppressor genes

Loss of function → ↑ cancer risk; both (**two**) alleles of a **tumor** suppressor gene must be lost for expression of disease (the Knudson 2-hit hypothesis).

GENE	GENE PRODUCT	ASSOCIATED CONDITION
APC	Negative regulator of β -catenin/WNT pathway	C olorectal cancer (associated with FAP)
BRCA1/BRCA2	BRCA1/BRCA2 proteins	BR east, ovarian, prostate, pancreatic C ancers
CDKN2A	p16, blocks $G_1 \rightarrow S$ phase	Many cancers (eg, melanoma, lung, pancreatic)
DCC	DCC — D eleted in C olorectal C ancer	Colorectal cancer
SMAD4 (DPC4)	DPC — D eleted in P ancreatic C ancer	Pancreatic cancer, colorectal cancer
MEN1	MEN in	M ultiple E ndocrine N eoplasia type 1
NF1	Neurofibromin (Ras GTPase activating protein)	N euro F ibromatosis type 1
NF2	Merlin (schwannomin) protein	N euro F ibromatosis type 2
PTEN	Negative regulator of PI3k/AKT pathway	P rostate, breasT , and EN dometrial cancers
RB1	Inhibits E2F; blocks $G_1 \rightarrow S$ phase	R etinoblastoma, osteosarcoma (B one cancer)
TP53	p53, activates p21, blocks $G_1 \rightarrow S$ phase	Most cancers, Li-Fraumeni (SBLA) syndrome (multiple malignancies at early age; S arcoma, B reast/ B rain, L ung/ L eukemia, A drenal gland)
TSC1	Hamartin protein	T uberous s clerosis complex
TSC2	Tu berin (“ 2 berin”)	Tuberous sclerosis complex
VHL	Inhibits hypoxia-inducible factor 1α	v on H ippel- L indau disease
WT1	Urogenital development transcription factor	W ilms T umor (nephroblastoma)

Carcinogens

TOXIN	EXPOSURE	ORGAN	IMPACT
Aflatoxins (<i>Aspergillus</i>)	Stored grains and nuts	Liver	Hepatocellular carcinoma
Alkylating agents	Oncologic chemotherapy	Blood	Leukemia/lymphoma
Aromatic amines (eg, benzidine, 2-naphthylamine)	Textile industry (dyes), tobacco smoke (2-naphthylamine)	Bladder	Transitional cell carcinoma
Arsenic	Herbicides (vineyard workers), metal smelting, wood preservation	Liver Lung Skin	Hepatic angiosarcoma Lung cancer Squamous cell carcinoma
Asbestos	Old roofing material, shipyard workers	Lung	Bronchogenic carcinoma > mesothelioma
Tobacco smoke		Bladder Cervix Esophagus Kidney Larynx Lung Oropharynx Pancreas	Transitional cell carcinoma Squamous cell carcinoma Squamous cell carcinoma/ adenocarcinoma Renal cell carcinoma Squamous cell carcinoma Squamous cell and small cell carcinoma Squamous cell carcinoma Pancreatic adenocarcinoma
Ethanol		Esophagus Liver Breast	Squamous cell carcinoma Hepatocellular carcinoma Breast cancer
Ionizing radiation		Blood Thyroid	Leukemia Papillary thyroid carcinoma
Nickel, chromium, beryllium, silica	Occupational exposure	Lung	Lung cancer
Nitrosamines	Smoked foods	Stomach	Gastric cancer (intestinal type)
Radon	Byproduct of uranium decay, accumulates in basements	Lung	Lung cancer (2nd leading cause after tobacco smoke)
Vinyl chloride	Used to make PVC pipes	Liver	Hepatic angiosarcoma

Oncogenic microbes

MICROBE	ASSOCIATED CANCER
EBV	Burkitt lymphoma, Hodgkin lymphoma, nasopharyngeal carcinoma, 1° CNS lymphoma (in immunocompromised patients)
HBV, HCV	Hepatocellular carcinoma
HHV-8	Kaposi (“Ka ^{oo} si”) sarcoma
HPV (usually types 16, 18)	Cervical and penile/anal carcinoma, head and neck cancer
<i>H pylori</i>	Gastric adenocarcinoma and MALT lymphoma
HTLV-1	Adult T-cell Leukemia/Lymphoma
Liver fluke (<i>Clonorchis sinensis</i>)	Cholangiocarcinoma
<i>Schistosoma haematobium</i>	Squamous cell bladder cancer

Serum tumor markers

Tumor markers should not be used as the 1° tool for cancer diagnosis or screening. They may be used to monitor tumor recurrence and response to therapy, but definitive diagnosis is made via biopsy. Some can be associated with non-neoplastic conditions.

MARKER	IMPORTANT ASSOCIATIONS	NOTES
Alkaline phosphatase	Metastases to bone or liver, Paget disease of bone, seminoma (PLAP).	Exclude hepatic origin by checking LFTs and GGT levels.
α-fetoprotein	Hepatocellular carcinoma, endodermal sinus (yolk sac) tumor, mixed germ cell tumor, ataxia-telangiectasia, neural tube defects.	Normally made by fetus. Transiently elevated in pregnancy. High levels associated with neural tube and abdominal wall defects, low levels associated with Down syndrome.
hCG	Hydatidiform moles and Choriocarcinomas (Gestational trophoblastic disease), testicular cancer, mixed germ cell tumor.	Produced by syncytiotrophoblasts of the placenta.
CA 15-3/CA 27-29	Breast cancer.	
CA 19-9	Pancreatic adenocarcinoma.	
CA 125	Epithelial ovarian cancer.	
Calcitonin	Medullary thyroid carcinoma (alone and in MEN2A, MEN2B).	Calcitonin.
CEA	Colorectal and pancreatic cancers. Minor associations: gastric, breast, and medullary thyroid carcinomas.	CarcinoEmbryonic Antigen. Very nonspecific.
Chromogranin	Neuroendocrine tumors.	
LDH	Testicular germ cell tumors, ovarian dysgerminoma, other cancers.	Can be used as an indicator of tumor burden.
Neuron-specific enolase	Neuroendocrine tumors (eg, small cell lung cancer, carcinoid tumor, neuroblastoma).	
PSA	Prostate cancer.	Prostate-Specific Antigen. Also elevated in BPH and prostatitis. Questionable risk/benefit for screening. Marker for recurrence after treatment.

Liquid biopsy Noninvasive diagnostic test in body fluid (eg, blood, urine) using circulating tumor DNA (ctDNA), RNA (ctRNA), or circulating tumor cells (CTCs) for cancer profiling (eg, *EGFR* in lung cancer), treatment monitoring, and post-treatment surveillance (eg, colorectal cancer). Aids in early metastatic detection and personalized therapy.

Important immunohistochemical stains Determine primary site of origin for metastatic tumors and characterize tumors that are difficult to classify. Can have prognostic and predictive value.

STAIN	TARGET	TUMORS IDENTIFIED
Chromogranin and synaptophysin	Neuroendocrine cells	Small cell carcinoma of the lung, carcinoid tumor, neuroblastoma
Cytokeratin	Epithelial cells	Epithelial tumors (eg, squamous cell carcinoma)
Desmin	Muscle	Muscle tumors (eg, rhabdomyosarcoma)
GFAP	NeuroGlia (eg, astrocytes, Schwann cells, oligodendrocytes)	Astrocytoma, Glioblastoma
Neurofilament	Neurons	Neuronal tumors (eg, neuroblastoma)
PSA	Prostatic epithelium	Prostate cancer
PECAM-1/CD-31	Endothelial cells	Vascular tumors (eg, angiosarcoma)
S-100	Neural crest cells	Melanoma, schwannoma, Langerhans cell histiocytosis
TRAP	Tartrate-resistant acid phosphatase	Hairy cell leukemia
Vimentin	Mesenchymal tissue (eg, fibroblasts, endothelial cells, macrophages)	Mesenchymal tumors (eg, sarcoma), but also many other tumors (eg, endometrial carcinoma, renal cell carcinoma, meningioma)

P-glycoprotein ATP-dependent efflux pump also called multidrug resistance protein 1 (MDR1). Expressed in some cancer cells to pump out toxins, including chemotherapeutic agents (one mechanism of ↓ responsiveness or resistance to chemotherapy over time).

Paraneoplastic syndromes

MANIFESTATION	DESCRIPTION/MECHANISM	MOST COMMONLY ASSOCIATED TUMOR(S)
Musculoskeletal and cutaneous		
Dermatomyositis	Progressive proximal muscle weakness, Gottron papules, heliotrope rash	Adenocarcinomas, especially ovarian
Acanthosis nigricans	Hyperpigmented velvety plaques in axilla and neck	Gastric adenocarcinoma and other visceral malignancies
Sign of Leser-Trélat	Sudden onset of multiple seborrheic keratoses	GI adenocarcinomas and other visceral malignancies
Hypertrophic osteoarthropathy	Abnormal proliferation of skin and bone at distal extremities → clubbing, arthralgia, joint effusions, periostosis of tubular bones	Adenocarcinoma of the lung
Endocrine		
Hypercalcemia	PTHrP ↑ 1,25-(OH) ₂ vitamin D ₃ (calcitriol)	SCa ²⁺ mous cell carcinomas of lung, head, and neck; renal, bladder, breast, and ovarian carcinomas Lymphoma
Cushing syndrome	↑ ACTH	Small cell lung cancer
Hyponatremia (SIADH)	↑ ADH	
Hematologic		
Polycythemia	↑ Erythropoietin Paraneoplastic rise to High hematocrit levels	Pheochromocytoma, renal cell carcinoma, HCC, hemangioblastoma, leiomyoma
Pure red cell aplasia	Anemia with low reticulocytes	Thymoma
Good syndrome	Hypogammaglobulinemia	
Trousseau syndrome	Migratory superficial thrombophlebitis	Adenocarcinomas, especially pancreatic
Nonbacterial thrombotic endocarditis	Deposition of sterile platelet thrombi on heart valves	
Neuromuscular		
Anti-NMDA receptor encephalitis	Psychiatric disturbance, memory deficits, seizures, dyskinesias, autonomic instability, language dysfunction	Ovarian teratoma
Opsoclonus-myoclonus ataxia syndrome	“Dancing eyes, dancing feet”	Neuroblastoma (children), small cell lung cancer (adults)
Paraneoplastic cerebellar degeneration	Antibodies against antigens in Purkinje cells	Small cell lung cancer (anti-Hu), gynecologic and breast cancers (anti-Yo), and Hodgkin lymphoma (anti-Tr)
Paraneoplastic encephalomyelitis	Antibodies against Hu antigens in neurons	Small cell lung cancer
Lambert-Eaton myasthenic syndrome	Antibodies against presynaptic (P/Q-type) Ca ²⁺ channels at NMJ	
Myasthenia gravis	Antibodies against postsynaptic ACh receptors at NMJ	Thymoma